M2201 - Introduction to Graph Theory

TD1

Modeling Exercises

1 The barrel.

We want to take 4 liters from a barrel, using two bottles of 3 liters and 5 liters, respectively. The only allowed operations are: (1) fill completely a bottle from the barrel, (2) transfer the content of one bottle to the other (without overflow), and (3) transfer the content of a bottle back in the barrel.
→ Draw a picture showing all the possible states and transitions of this system. The solution should emerge directly from the picture.

2 Crossing the river with a wolf, a goat, and a cabbage.

A farmer needs to bring a wolf, a goat, and a cabbage across the river. However, his boat is tiny and can only carry two passengers (including the farmer). If he leaves the wolf and the goat alone together, the wolf will eat the goat. If he leaves the goat and the cabbage alone, the goat will eat the cabbage. How can he bring all three safely across the river?
→ Again, the objective is not "just" to solve the problem, but to draw its states and transitions. First, what is an appropriate representation of a state in this system? The solution must emerge directly from the picture. (For conciseness, we will use the letters `F`,`W`,`G`,`C` to denotes the characters.)

3 The Fan-Tan game.

We consider the following two player game: there are two rows of sticks, for example two identical rows with 3 sticks each (in general, these numbers could be different). In each turn, the current player can remove either one or two matches from a single row. The player who removes the last stick loses.

→ Is there a winning strategy for the first player? If yes, which one? (A "winning strategy" is a way to play that guarantees success.) Again, you must draw the states and transitions of this game.

4 Fort-Boyard (version 1).

There is a single row of 20 sticks. In each turn, the current player can remove 1, 2, or 3 consecutive sticks *at one extremity* of the row. The player who removes the last stick loses.

→ Draw the states and transitions. Do you prefer playing first or second? Why?

→ What if the winner is the one taking the last stick?

5 Fort-Boyard (version 2).

There is a single row of 5 sticks. In each turn, the current player can remove 1, 2 ou 3 consecutive sticks, but this time *anywhere* in the row. The player taking the last stick loses. Draw the configurations. Do you want to play first? Why?

6 Treasure Hunt.

Yet another two player game. We now consider a 3×3 board. Initially, the pirate (P) is located at the center. In each turn, the current player can move the pirate either vertically or horizontally to a non-empty cell to obtain the corresponding amount of money (fixed initially). If a player cannot make such a move, the game is over and the winner is the one with the largest amount of money. In case of ties, the second player wins.

1. Draw the configurations (on a large sheet of paper)
2. Do you prefer to play first or second?

20	10	20
	P	10
	10	

Do you know an algorithm that finds a winning strategy if there is one?